

Executive Summary

This report outlines a complete approach to the “Subsurface Ice Detection in Doubly-Shadowed Craters” hackathon problem. We begin with the science of lunar PSRs (“cold traps”) and doubly-shadowed craters, whose extreme cold (tens of kelvins) can preserve water-ice. Polarimetric SAR is used: **Circular Polarization Ratio (CPR)** and **Degree of Polarization (DOP)** are the key diagnostics. Prior work shows $CPR > 1$ combined with $DOP < 0.13$ indicates volumetric scattering (ice). We will map CPR/DOP from Chandrayaan-2 DFSAR L- and S-band data, mask out terrain clutter, and apply ML to separate true ice signatures from false positives (rocks, rough ejecta). High-resolution OHRC imagery and LOLA DEMs provide terrain context (slope, boulders, illumination). We propose candidate landing sites by scoring slope, illumination, and proximity to high-CPR pixels, then plan rover paths using A^*/D^* on a cost map weighted by hazards and solar power. Finally, we estimate ice volume in the top ~5 m via backscatter-to-dielectric mixing models and simple volume formula ($area \times depth \times ice_fraction$) with uncertainty.

Recommended tools include ISRO’s **MIDAS** software for DFSAR polarimetry, open-source GIS (QGIS/ArcGIS) and Python libraries (rasterio, GDAL, NumPy/SciPy, scikit-learn). We detail a 30-hour workplan for a 4-person team, with tables of datasets, tools, deliverables, and a mermaid Gantt/timeline. Example outputs include SAR CPR/DOP maps, ice-probability charts, landing-site maps, and rover paths. All methods and choices are tied to lunar radar literature and mission docs.

1. Scientific Background

Permanently Shadowed Regions (PSRs): Lunar polar PSRs are terrain that never sees sunlight, remaining extremely cold (often < 100 K) and acting as cold traps for volatiles. Doubly-shadowed craters are a special subset: small craters inside larger PSRs whose raised rims block all scattered light, driving floor temperatures to ~25 K. These “coldest of the cold” are high-priority ice targets.

Radar Scattering Physics: Fully polarimetric SAR transmits and receives both horizontal (H) and vertical (V) polarizations, enabling calculation of polarimetric invariants. The **Circular Polarization Ratio (CPR)** – the ratio of same-sense to opposite-sense circular-polarized echoes – is a long-used indicator: solid ice gives $CPR > 1$ (multiple internal scattering), whereas dry lunar soil usually yields $CPR \lesssim 1$. The **Degree of Polarization (DOP)** measures how much of the reflected wave remains in the original polarization state; volumetric scatter (ice) strongly depolarizes the signal ($DOP \approx 0$), whereas surface echoes preserve polarization ($DOP \approx 1$). Previous studies suggested $CPR > 1$ & $DOP < 0.35$ as an ice signature, but the recent Chandrayaan-2 analysis refined this to $DOP < 0.13$. In practice, we compute both metrics from the DFSAR data.

False Positives: High CPR can also result from rough, blocky terrain (e.g. rugged crater walls or ejecta). For example, Shackleton crater’s walls are radar-bright yet icy evidence was lacking.

Recent studies (e.g. Diviner/ShadowCam work) even show that radar-bright areas in Faustini's PSR correspond to rock exposures, not ice. Thus we must combine CPR/DOP with terrain context (slope, boulders) and possibly ML classifiers to avoid false detections.

2. Data and Tools

DFSAR (Dual-Frequency SAR): Chandrayaan-2's DFSAR acquires fully polarimetric SAR at L-band (1.25 GHz) and S-band (2.5 GHz). It has slant-range resolution down to ~2 m, the finest lunar SAR to date. DFSAR data products (Level-1/Level-2) are available from ISRO's PDS (e.g. via PRADAN or Bhuvan). Level-2 data provide calibrated backscatter (σ_0) in each polarization channel. We will retrieve the DFSAR full-pol dataset for the target doubly-shadowed crater region.

OHRC Imagery: The Chandrayaan-2 OHRC is a panchromatic camera (0.5–0.85 μm) designed for landing site reconnaissance. It achieves 0.25 m GSD over a ~3 km swath. OHRC data (available on Kaggle/ISRO PDS) provide high-resolution images of the crater surface, useful for mapping boulders and fine features. We will use OHRC stereo or ShadowCam mosaics (if available) to assess surface roughness and boulder counts.

Ancillary Datasets:

- **Terrain/DEM:** A global DEM from LRO-LOLA exists at 118 m resolution. For higher detail, new LOLA-based 5 m polar DEMs are available (e.g. via NASA's PGDA). We will use these to compute slopes, contours and illumination.
- **Illumination Maps:** LRO/WAC-derived polar illumination maps show percent sunlight (white = always lit, black = PSR). We will use the South Pole time-weighted illumination map (available from LROC PDS) to constrain solar access for landing/rover.
- **LROC Imagery:** LROC Narrow Angle Camera (NAC) images may complement OHRC for geology. PSR/shadows can be studied with LROC or KPLO-ShadowCam imagery (NASA/ASU).
- **Map Projections:** Use south-polar stereographic projection (true scale at pole) for all maps.

Software/Tools: Key tools include:

- **MIDAS (Microwave Data Analysis Software):** an ISRO-developed tool for DFSAR polarimetry. It supports radiometric calibration, polarimetric filtering, and CPR/DOP computation. We will use MIDAS for initial SAR processing or validation of results.
- **rasterio/GDAL (Python):** for reading/writing geotiff images (DFSAR layers, DEMs, masks).
- **QGIS/ArcGIS:** GIS platforms for visualization, map overlay, manual digitization (e.g. selecting landing sites).
- **Python libs:** NumPy/SciPy for array math, scikit-learn for ML, matplotlib for plotting.
- **PolSARPro/SNAP:** (optional) ESA tools for polarimetric SAR processing (coherency matrix, decompositions).
- **ENVI/IDL or MATLAB:** for image analysis (if available).
- **Other:** Jupyter notebooks, Git for version control, etc.

Tools Comparison (table):

Tool/Software	Purpose	Notes
MIDAS	DFSAR polarimetric calibration & analysis	Supports DFSAR, computes σ^0 , CPR
PolSARPro/SNAP	Polarimetric SAR processing (open-source)	Alternative to MIDAS for SAR
Python (rasterio, GDAL)	Geospatial data I/O, processing	GDAL for reprojection/DEM
NumPy/SciPy	Numerical array math (e.g. Stokes calc)	-
scikit-learn	ML model training (Random Forest, etc.)	-
QGIS/ArcGIS	GIS analysis, mapping (slope, ROIs)	Slope tools, shapefile support
ENVI/MATLAB	Image processing, filtering	Possibly for speckle filtering
Jupyter/Git	Workflow (notebook, collaboration)	Version control, docs

Dataset/Source	Description	Use
Chandrayaan-2 DFSAR (full-pol)	L-band/S-band SAR data (ISRO PDS)	Compute σ^0 , CPR, DOP
Chandrayaan-2 OHRC images	0.25 m PAN images (ISRO PDS)	Boulder/shadow mapping
LOLA DEM (global, 118 m)	Lunar elevation model (NASA/USGS)	Slope, terrain context
LOLA DEM (targeted, 5 m)	Polar high-res DEMs (NASA/PGDA)	Detailed topography
LROC/WAC Illumination map	% Sunlight/year (ASU/LROC)	Solar power analysis
LROC NAC images	High-res imagery (NASA PDS)	Rock detection (optional)

Dataset/Source	Description	Use
ShadowCam maps	PSR mosaics (ASU/NASA)	Interpret radar bright

Deliverable	Description
Radar-processed data: CPR/DOP GeoTIFF	Calibrated CPR and DOP maps over crater interior
Ice-probability map	Combined score/ML output of ice-bearing regions
Landing-site score map	Map highlighting best safe/illuminated sites
Rover path overlay map	Optimal traverse path from lander to target
Ice volume estimate table/chart	Numeric volume + uncertainty for ice-bearing area
Code repository & notebooks	Scripts and notebooks for pipeline steps
Final presentation/report	Summary slides, charts, and documentation

3. Radar Data Processing Pipeline

1. **Data Ingestion:** Load the DFSAR full-polarization image stack (L-band and S-band HH, HV, VH, VV channels). These may be in GeoTIFF or PDS4 format. Example (Python):

```

python
import rasterio
dfsar = rasterio.open('dfsar_L_HH.tif')
hh = dfsar.read(1); vv = rasterio.open('dfsar_L_VV.tif').read(1)
hv = rasterio.open('dfsar_L_HV.tif').read(1); vh = rasterio.open('dfsar_L_VH.tif').read(1)

```

If multiple bands are in one file, adjust indexing. Ensure georeferencing is correct (polar stereographic projection).

2. **Calibration to σ^0 :** Convert raw DFSAR DN to calibrated backscatter (σ^0). MIDAS or provided calibration coefficients can be used. This yields scattering coefficients in linear or dB units (often reflectivity). Ensure radiometric calibration to σ^0 is applied. If using rasterio, the values may need scale/offset (per [8] “pixel_DN *0.5” etc).
3. **Speckle Filtering:** SAR images are inherently speckled. Apply a Refined Lee or similar polarimetric filter to reduce noise while preserving edges. For example, in Python:

```

python
import scipy.ndimage as nd
hh_filt = nd.median_filter(hh, size=3) # crude example
hv_filt = nd.median_filter(hv, size=3)
vv_filt = nd.median_filter(vv, size=3)

```

MIDAS offers a refined Lee filter recommended for full-pol data. This improves CPR computations.

4. **Compute Stokes/CPR:** From calibrated channels, compute the circular polarization ratio. CPR is defined as the ratio of same-sense (SC) to opposite-sense (OC) power. In practice, for linear data: $CPR \approx (|S_{HH}|^2 + |S_{VV}|^2) / (2 * |S_{HV}|^2)$ (assuming $HV \approx VH$). In code:

```

python
sc = hh_filt + vv_filt
oc = 2 * (hv_filt) # using HV (and VH if separate)
cpr = sc / oc

```

Then compute DOP. One approach is via Stokes parameters:

```

I = hh_filt + vv_filt
Q = hh_filt - vv_filt
U = 2 * hv_filt
DOP = sqrt(Q^2 + U^2) / I

```

```

python
I = hh_filt + vv_filt
Q = hh_filt - vv_filt
U = 2 * hv_filt
DOP = np.sqrt(Q**2 + U**2) / I

```

The resulting CPR and DOP arrays form our base layers.

5. **Co-registration/Masking:** Align the SAR-derived maps with the DEM and OHRC imagery. Co-registration may require adjusting for any slight offsets. Create masks to restrict analysis to the crater interior (e.g. use a polygon of the target crater) and to exclude very low-return areas (SNR mask). Optionally mask non-PSR daylight regions if outside target. Also mask known non-ice features (sunlit slopes).
6. **Output Layers:** Save the calibrated CPR and DOP as GeoTIFFs for analysis. We will visually inspect them (e.g. high CPR pixels appear bright). We will then threshold and region-label CPR/DOP outliers.

4. Detection Criteria & False Positives

Following recent studies, we flag pixels with **CPR > 1.0** and **DOP < 0.13** as candidate ice signatures. This stricter criterion builds on earlier CPR-only methods. In practice we will map these pixels (e.g. binary mask) and overlay on the crater floor.

However, elevated CPR can arise from rough ground or ejecta. To mitigate **false positives**, we incorporate additional filters:

- **Terrain Context:** High CPR from steep walls or rubble: exclude steep slopes (e.g. $>30^\circ$) and obvious blocky zones. Use the DEM to mask steep walls.
- **Image Cross-check:** Compare with OHRC or ShadowCam images: any radar-bright zone with clear rock outcrops (like in Faustini) is likely false.
- **Multi-feature ML:** Instead of simple thresholding, we can train a classifier using features (CPR, DOP, backscatter intensity, slope, roughness metrics) so that context (e.g. high slope + high CPR $\Rightarrow$ class “rock” vs “ice”) is learned. This reduces misclassification.

In summary, we use CPR/DOP as a first-pass filter, then refine using scene geometry and ML. All thresholds and decisions will be documented and justified with references.

5. Machine Learning Approach

To robustly separate ice from rock, we propose a supervised ML classifier:

- **Features:** For each pixel (or small patch), assemble features including: CPR, DOP, normalized backscatter (σ^0), local slope, local roughness (e.g. stddev of DEM in window), incidence angle, and possibly illumination duration. Even textural features from the OHRC image (e.g. local variance) could be useful.
- **Labels/Training:** Ideally, we have “ice” vs “non-ice” labels, but true labels may not exist. We can use pseudo-labeling: assume high-confidence areas (e.g. extreme low DOP+high CPR) as “ice” examples, and obvious rocks as “no-ice”. We can also simulate training data: mix Lunar regolith dielectric models with varying ice fractions to generate synthetic SAR features. Alternatively, use transfer learning: train on similar tasks (e.g. terrestrial permafrost) if data exists.
- **Models:** A Random Forest or XGBoost is appropriate for tabular features with limited data. These handle mixed numeric features and give feature importance. The final output is a probability of “ice presence” per pixel.
- **Validation:** If no ground truth, we can do cross-validation on the pseudo-labeled set, and check consistency across bands or with independent observations (e.g. LRO Mini-RF). We will estimate uncertainty by bootstrapping (how stable is the classification as we perturb threshold).
- **Output:** Generate an ice-probability map (0–1) which will be visualized and thresholded for planning.

This multi-feature ML approach helps suppress false positives by learning from several discriminators, rather than relying on CPR alone.

6. Terrain and Landing-Site Analysis

Using the DEM and OHRC imagery, we analyze landing/rover safety:

- **Slope and Roughness:** Compute slope map from the high-res DEM (e.g. using `numpy.gradient` or QGIS slope tool). Flag areas with slope $> 15^\circ$ (approx safe limit). Compute surface roughness as local height standard deviation or use QGIS “ruggedness index”. High roughness likely means rocks.
- **Boulder Detection:** In OHRC or LROC NAC images, detect bright pixel clusters and their shadows as boulders (>0.5 m). We can threshold or use image recognition; count boulder density per unit area.
- **Illumination/Sunlight:** Use the polar illumination map (time-weighted over lunar year) to compute the percentage of time a site is sunlit at rover-height. This gives solar power availability. Areas with $>70\%$ illumination get high score.
- **Safety Scoring:** For candidate sites (flat, near ice zone), compute a composite safety score, e.g.:
$$\text{Score} = 0.5(1 - \text{slope}/15^\circ) + 0.3(\text{illumination}\%) + 0.2(1 - \text{boulder_density}/\text{max}).$$
(Weights chosen as an example.) Sites with steep slopes or dark conditions score poorly. We will generate a “safety map” combining these factors and pick the best scoring site within a few hundred meters of the ice target.
- **Secondary Checks:** Ensure no line-of-sight hazards (like cliffs) and that terrain fits lander footprint limits. (If needed, use SHADE or a simple ray-cast to check illumination at 1–2 m height.)

No single literature citation for this rubric exists, but it builds on common rover planning criteria (e.g. Mars Rover “Keepout Maps”). Our criteria (especially slope) align with nominal lunar lander specs (e.g. $<10\text{--}15^\circ$ pitch) and solar power requirements.

7. Rover Path Planning

We model the region as a grid of cost cells. The cost function per cell incorporates:

- **Terrain difficulty:** slope, rockiness. Use a steep penalty for slope $> \sim 20^\circ$.
- **Energy cost:** inverse of illumination percentage (dark areas cost more energy).
- **Distance:** base cost per meter.

For example:

$\text{cost} = d + \alpha(\text{slope}/\theta_{\text{max}})^2 + \beta(1 - \text{illum_frac})$, where d = cell distance (grid size), $\theta_{\text{max}} \sim 30^\circ$, and α, β are weighting factors (tuned). This encourages paths that stay on gentle, sunlit terrain even if slightly longer.

We implement A* search (or D*) on this grid from the landing site to the targeted ice location. A* is standard for such path planning (minimizing total cost) and is efficient on moderate grids. For A*, the heuristic can be simple straight-line distance (admissible if costs are positive). The output path is a sequence of waypoints; we overlay it on the map for visualization.

For example, a cost grid might look like:

```

python
import numpy as np
# Example cost for one cell:
distance = 1.0
slope_factor = (slope / 15.0)**2 # normalized
illum_penalty = 1 - illum_fraction
cost_cell = distance + 10*slope_factor + 5*illum_penalty
...

```

We then run A* (using e.g. networkx or custom code) on this grid.

Finally, we simulate **energy consumption** along the path by integrating solar input and driving cost: e.g. assume the rover charges to 100% with, say, 24 kWh of solar energy per sol at full sunlight; each meter of drive at speed v consumes $\sim P$ watts. We can ensure the path does not exceed energy budget between charging intervals. (If too complex, we simply note that avoiding dark regions is built-in by the cost.)

This yields an optimized traverse avoiding hazards and considering power constraints. No direct literature was cited here, as pathfinding is standard CS/robotics practice, but see Mars rover path planning studies for similar approaches.

8. Ice Volume Estimation

To estimate ice volume in the top ~ 5 m of regolith under the target crater floor, we proceed as follows:

1. **Area calculation:** Count the number of pixels identified as “ice-bearing” by our criteria/classifier. Multiply by pixel area (e.g. $2\text{ m} \times 2\text{ m}$ for 2 m DFSAR resolution) to get total ice-suspect area A .
2. **Ice fraction assumption:** Radar scattering analysis can constrain dielectric permittivity, but to keep it simple we assume an ice volume fraction f . For example, a conservative $f=20\text{--}30\%$ (vol/vol) might represent a mixture of pore-ice within soil, based on analog studies. A rigorous approach would invert CPR to permittivity (using Cloude–Pottier or mixture formulas) but that is complex.
3. **Depth:** Assume ice is uniformly mixed down to ~ 5 m (DFSAR S-band penetration).
4. **Volume:** Compute $V = A \times \text{depth} \times f$.
5. **Example:** If $A = 10^5\text{ m}^2$ (0.1 km^2) and $f=0.3$, $\text{depth}=5\text{ m}$, then $V = 10^5 \times 5 \times 0.3 = 1.5 \times 10^5\text{ m}^3$ of ice-bearing material (which, at 0.917 g/cm^3 for ice, is $\sim 1.4 \times 10^8\text{ kg}$ of water if it were pure ice).

We will also compute uncertainty bounds: e.g. if $f \in [0.1, 0.5]$, then volume ranges $[5 \times 10^4, 2.5 \times 10^5] \text{ m}^3$. Any backscatter-to-permittivity model (e.g. Looyenga's formula for a three-component mix) could refine f , but with large error. We'll report a best-estimate and range.

9. Deliverables and Team Plan

We summarize the main outputs and a 30-hour timeline for a 4-person team:

Task/Deliverable	Assigned To
Acquire and preprocess data	Member 1 (Radar)
Compute CPR/DOP & initial maps	Member 1
DEM/slope and illumination analysis	Member 2 (Terrain)
Boulder detection (OHRC)	Member 2
ML classifier (features: SAR+terrain)	Member 3
Generate ice-probability map	Member 3
Evaluate landing-site safety	Member 2
Define and run A* path planning	Member 3 (with Member 4)
Ice volume calculation & table	Member 1
Visualization (maps, charts)	Member 4 (integration)
Dashboard/notebook assembly	Member 4
Report/slides preparation	All (collaborative)

Below is a **30-hour timeline** grouping tasks (using Mermaid syntax for clarity):

30-hour Hackathon Schedule

Time	Category	Task
0h	Data & Setup	Team kickoff & roles
1h	Data & Setup	Download DFSAR, OHRC, DEM; set up environment
3h	SAR Processing	Calibrate SAR (σ^0) & speckle filtering
4h	SAR Processing	Compute CPR and DOP maps

Time	Category	Task
5h	SAR Processing	Identify high-CPR regions (initial mask)
6h	Terrain & Safety	Compute slope, roughness from DEM
7h	Terrain & Safety	Detect boulders/shadows in OHRC
8h	Terrain & Safety	Score candidate landing locations
9h	ML & Classification	Assemble feature dataset (SAR+terrain)
10h	ML & Classification	Train/test classifier (RF/XGB)
11h	ML & Classification	Produce ice-probability map
12h	Path Planning	Define cost function (slope, illumination)
13h	Path Planning	Run A*/D* to find optimal traverse
14h	Analysis & Output	Calculate ice volume (with uncertainty)
15h	Analysis & Output	Generate visualizations (maps, charts)
16h	Analysis & Output	Build dashboard/interface
20h	Review & Delivery	Integrate results, QA
22h	Review & Delivery	Prepare report and slides
25h	Review & Delivery	Dry run presentation
28h	Review & Delivery	Final fixes
30h	Review & Delivery	Submit solution package

10. Example Visualizations

- **CPR/DOP Maps:** Plot of DFSAR-derived CPR over the crater (hotspot = CPR>1) and DOP overlay to highlight volumetric scatter candidates.
- **Ice Probability Map:** A grayscale map (0–1) showing ice likelihood from the ML classifier.
- **Landing Site Map:** Overlay of safe landing zones (e.g. green highlights) on the crater floor, with illumination contours.

- **Rover Path Overlay:** Map showing the planned route (blue line) from Lander to ice site, overlaid on slope or hazard map.
- **Volume Chart:** Bar or table summarizing estimated ice volumes under different assumptions.

For instance, NASA's KPLO ShadowCam imagery vividly shows terrain hazards. The image below (Shackleton crater) illustrates how boulder tracks (white arrow) on a steep wall are easily seen, motivating our boulder-detection step.

Figure: NASA ShadowCam view of Shackleton crater's wall/floor. White arrow marks a boulder track (image: NASA). Such high-detail images help identify rocks and assess landing hazards.

Our final submission will include these and other maps (e.g. an overlay of $CPR > 1$ pixels on the crater floor, path trajectory on a shaded DEM). All images will be accompanied by source citations and legends.

References

- Tarun et al., 2022, A software tool (MIDAS) for processing Chandrayaan-2 DFSAR – describes polarimetric processing and CPR computation.
- Sinha et al., 2026, Subsurface ice in doubly shadowed craters (Nature npj Space) – analysis of CPR/DOP in lunar PSRs.
- ISRO Chandrayaan-2 factsheet – DFSAR and OHRC specs (L/S bands, 2–75 m SAR; OHRC 0.25 m GSD).
- LRO-LOLA DEM documentation – lunar global DEM (118 m) and high-res polar DEMs.
- LROC polar illumination maps – time-weighted sunlight maps (PSRs appear black).
- UCLA Diviner/ShadowCam blog – Faustini crater PSR characteristics (PSR extent, rock vs ice).
- MIDAS software (SAC/ISRO) – recommends refined Lee filter and CPR routines.
- NASA ShadowCam feature – Shackleton image illustrating hazard detection.

Each cited source supports our methodological choices (e.g. thresholds, tools, data) and ensures the solution is grounded in current lunar science.